

Machine Learning

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Supervised Learning

} It's a statistical modelling technique
The given dataset will have features and label.

Regression *

Features: any kind of data

Label: Continuous ND.

Linear Regression

Classification

Features: any kind of data

Label: Categorical Data

or
Discrete ND.

Inferential stat modelling process

Assuming the given dataset is COMPATIBLE for ML algo.

① Data must be COMPLETE

② Data must be STRICTLY NUMERIC

} during descriptive stats.

① Separate data as features and label (represent the data in the form of numpy array)

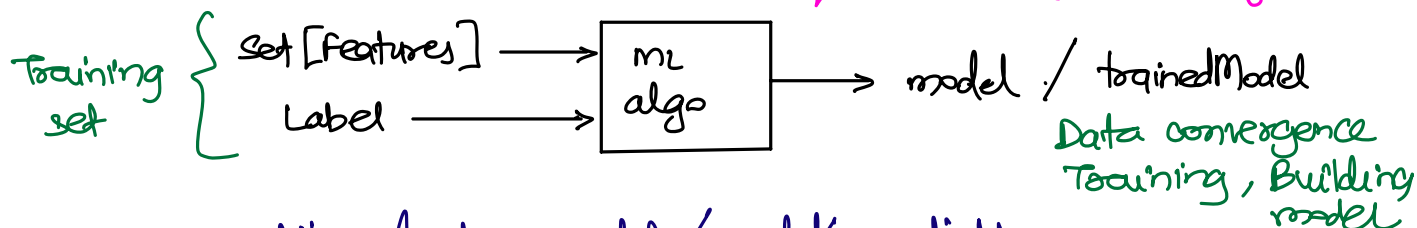
② Split the data into Training set and Testing set. `train_test_split`

`x_train, y_train`
Training set → used to train the model (Data Convergence)

`x_test, y_test`
Testing set → used to check the quality of the model.

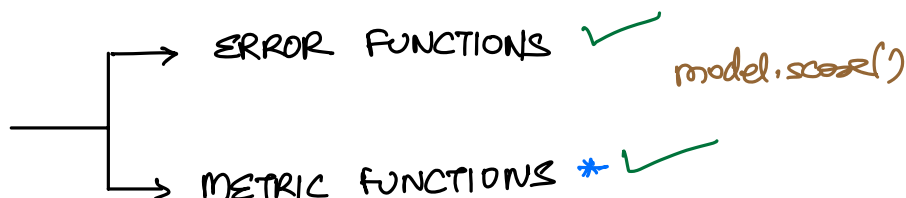
③ Select the relevant ML algorithms and start the training process.

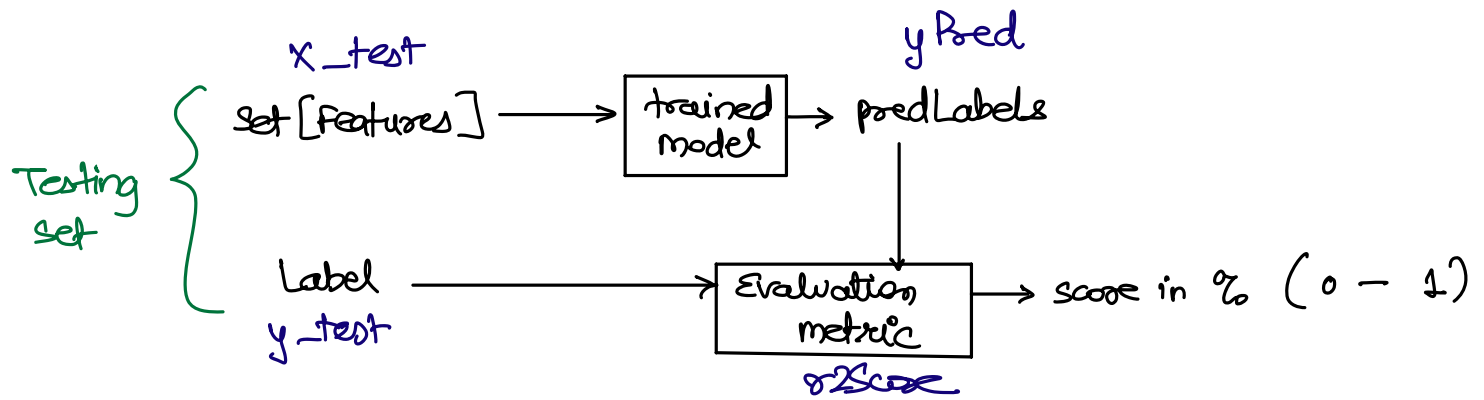
⇒ `model.fit(x_train, y_train)`



④ Check the quality of the model / model's prediction.

Goal: check the quality of the model w.r.t prediction





⑤ Decide whether to Approve model or Improve model.

if testScore > trainScore & testScore >= CL:

Approve the model and deploy the same

else:

Improve model and recheck the quality.

Generalization:

- or -

Generalized Model

Regression
Evaluation

Error functions

Less the error, better the model

① MSE = Mean Squared Error

② MAE = Mean Absolute Error

Metric Function

- If value near to 1, better the model
- If value greater than equal to CL, approve model

Adjusted R^2 Score

Feature Engineering (FE)

Feature engineering is applicable for **Supervised Learning** only.

FE. is applied on **INDEPENDENT VARIABLES**.
(feature cols)

Goal:

The ideal goal of FE is to

SELECT/SHORTLIST the set of **SIGNIFICANT FEATURES**

STATISTICALLY SIGNIFICANT FEATURES

based on:

(a) Label column

(b) ML algorithm

(c) **Statistical Testing formula.**

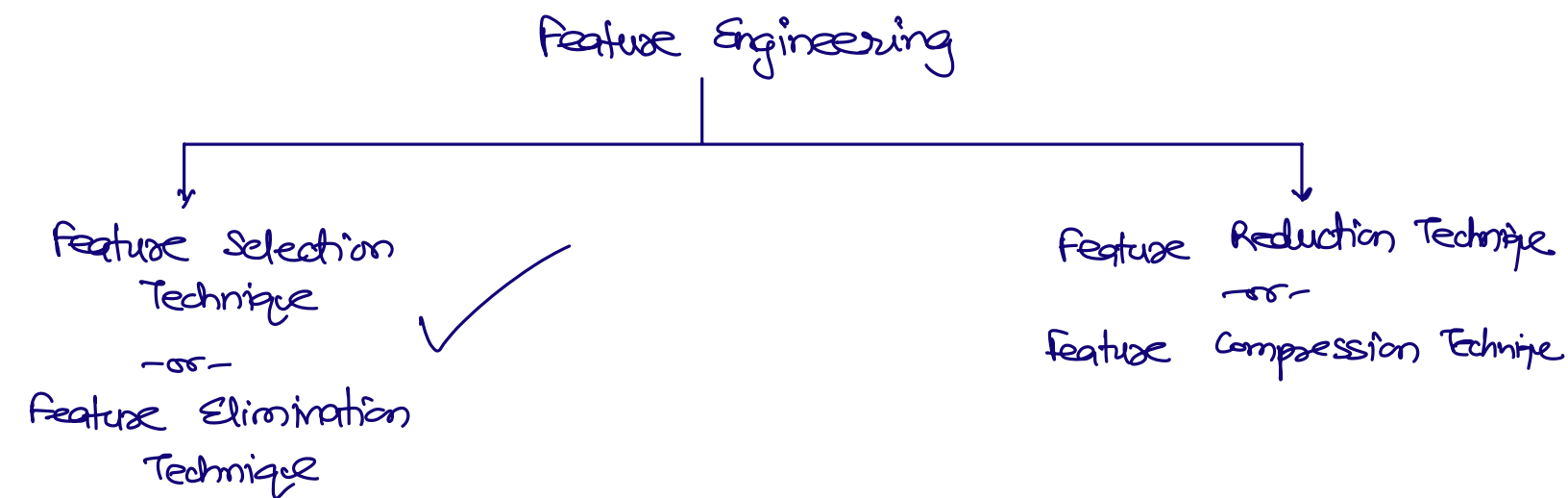
if feature cols is Normally Dist.:

→ Parametric Test

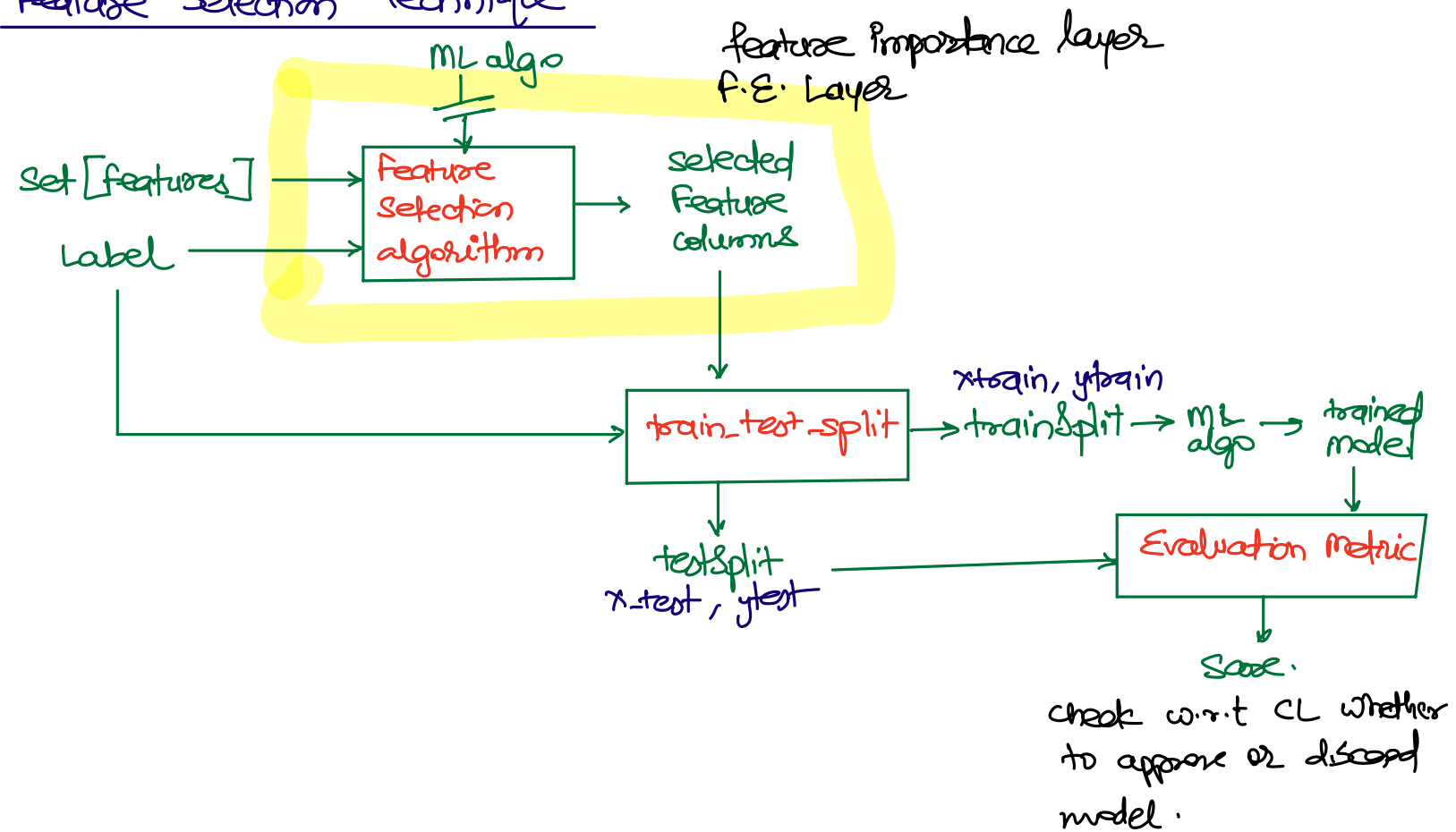
else:

→ Non-parametric Test

so that you can **ACHIEVE AN OPTIMAL MODEL**.



Feature Selection Technique



List of Feature Selection algo

- ① Correlation Analysis
- ② Backward Elimination Technique (OLS)
- ③ Recursive Feature Elimination (RFE)
- ④ Select from Model (SFM)

} → Applicable for Regression use-cases only.

} → Applicable for BOTH Regression & Classification use-cases.

Backward Elimination using OLS (Ordinary Least Square)

just another Regression algo. 😊

① Perform ALL-IN } → Include all features of co-eff and intercept.

$$\text{profit} = b_0(1) + b_1[\text{California}] + b_2[\text{Florida}] + b_3[\text{NY}] + b_4[\text{adSpend}] + b_5[\text{admSpend}] + b_6[\text{markSpend}]$$

coeff

OLS formula expects you to supply FULL input*

coeff features →

California
Florida
NY
adSpend
admSpend
markSpend

intercept features → 1*

② Decide / get the SL of the project.

③ Perform OLS operation on the given feature set and get the pValue of FULL input*

④ Select the feature column that has the highest p-value.

⑤ Check the following condition

if pvalue > SL:

eliminate that feature col
recreate new feature set.*

else:

go to step ⑦

⑥ Repeat from step 2 with new feature setⁿ

⑦ Consider the given feature set as final feature set for modelling.